Naive Bayes Foundations: A Bayes' Theorem Problem

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Bayes' theorem.

1. Overview

Key point: Three machines make markers; a random marker turns out defective. Bayes' theorem, with the total probability rule for the evidence, gives the chance it came from each machine.

This Note works through a classic problem with Bayes' theorem. The Note also introduces the **law of total probability** (G-1053), the usual way to compute the evidence $P(B)$ in the denominator (the overall probability of what we observed, here a defective marker). Naive Bayes computes its evidence the same way.

2. The problem

Key point: Production shares 20%, 30%, 50%; defect rates 5%, 3%, 1%.

A factory makes markers on three machines:

Machine	Share of all markers	Defect rate of that machine
M1	20%	5%
M2	30%	3%
M3	50%	1%

A marker is picked at random from a batch and turns out to be **defective**. What is the probability that it was made by M3?

Figure 1 shows the idea before any formula. In a typical batch of 1000 markers, M1, M2 and M3 make 200, 300 and 500. Their defect rates leave 10, 9 and 5 defective markers. Knowing the picked marker is defective means it is one of these 24, and only 5 of the 24 came from M3.

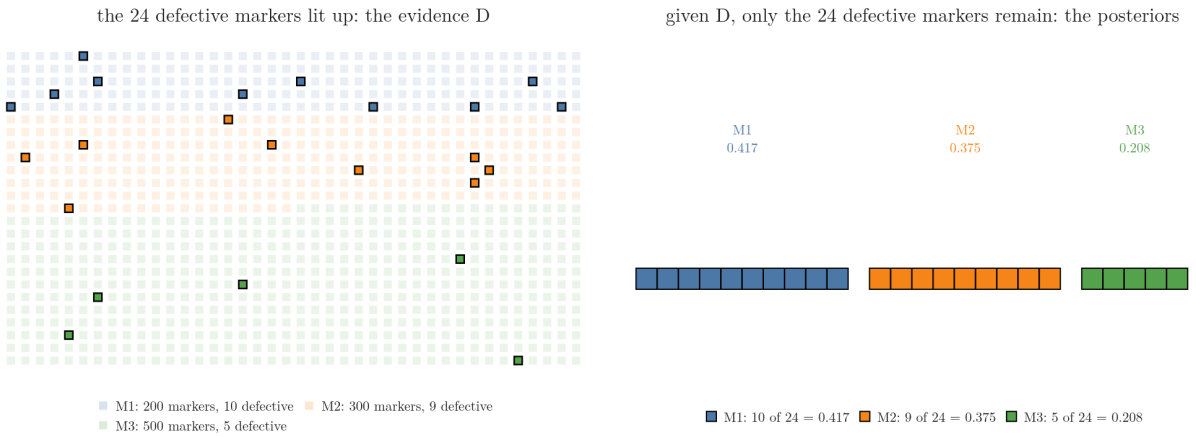


Figure 1: A batch of 1000 markers, one square each, coloured by machine. The 24 defective ones light up (10 from M1, 9 from M2, 5 from M3); given that the marker is defective, only those 24 remain, and M3 supplies 5 of 24 = 0.208.

3. Writing down what we know

> **Key point:** The production shares are priors P(M); the defect rates are likelihoods P(D | M).

Let D be "the marker is defective". In probability language:

- **Priors** (G-1565): $P(M1) = 0.2$, $P(M2) = 0.3$, $P(M3) = 0.5$.
- **Likelihoods** (G-1086): $P(D \mid M1) = 0.05$, $P(D \mid M2) = 0.03$, $P(D \mid M3) = 0.01$.

The question asks for $P(M3 \mid D)$. By **Bayes' theorem** (G-269; it turns $P(D \mid M3)$, the chance of a defect given the machine, into $P(M3 \mid D)$, the chance of the machine given a defect; see the names of the four parts),

$$P(M3 \mid D) = \frac{P(D \mid M3) \times P(M3)}{P(D)}$$

Everything on the right is known except $P(D)$, the overall probability that a marker is defective.

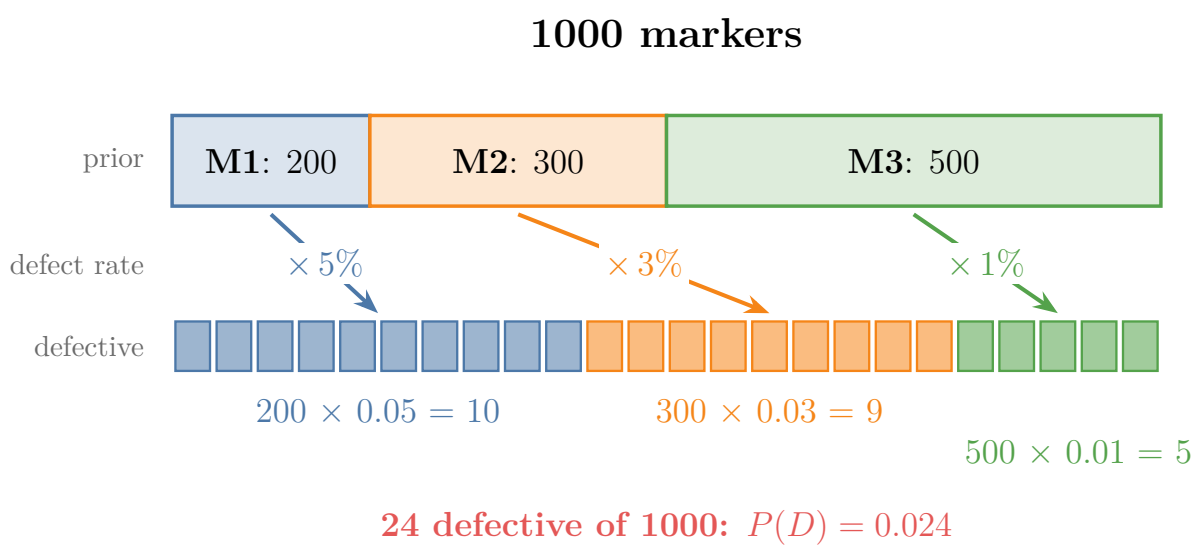


Figure 2: The problem as counts in a batch of 1000 markers. Top: the priors, with widths to scale. Bottom: one square per defective marker, from each machine's count times its defect rate

Figure 2 turns the priors and likelihoods into expected counts for 1000 markers. Watch the bottom row: M3 makes the most markers but supplies the fewest defective squares.

4. The evidence: total probability

> **Key point:** A defective marker came from exactly one machine, so P(D) is the sum of the three "defective and from machine i" probabilities.

A defective marker must have come from M1, M2 or M3, and from only one of them. These three cases are mutually exclusive (they cannot happen together), so their probabilities add. The symbol $\cap$ means "and": $D \cap M1$ is "defective and made by M1".

$$\begin{aligned} P(D) &= P(D \cap M1) \\ &+ P(D \cap M2) \\ &+ P(D \cap M3) \end{aligned}$$

Each term is a **likelihood** (G-1086) times a prior, by this rule from the proof of Bayes' theorem:

$$P(A \cap B) = P(B \mid A)\, P(A)$$

So for every machine,

$$P(D \cap M) = P(D \mid M)\, P(M)$$

and the sum becomes

$$\begin{aligned} P(D) = {} & P(D \mid M1)P(M1) \\ & + P(D \mid M2)P(M2) \\ & + P(D \mid M3)P(M3) \end{aligned}$$

This sum is the **law of total probability** (G-1053). With numbers:

$$\begin{aligned} P(D) = {} & 0.05 \times 0.2 \\ & + 0.03 \times 0.3 \\ & + 0.01 \times 0.5 \end{aligned}$$

$$P(D) = 0.010 + 0.009 + 0.005$$

$$P(D) = 0.024$$

So 2.4% of all markers are defective. Figure 3 shows the same calculation as a **probability tree** (G-1573). Each path from the left to a defective marker passes through one machine, and multiplying the numbers along the path gives that machine's term of the sum above.

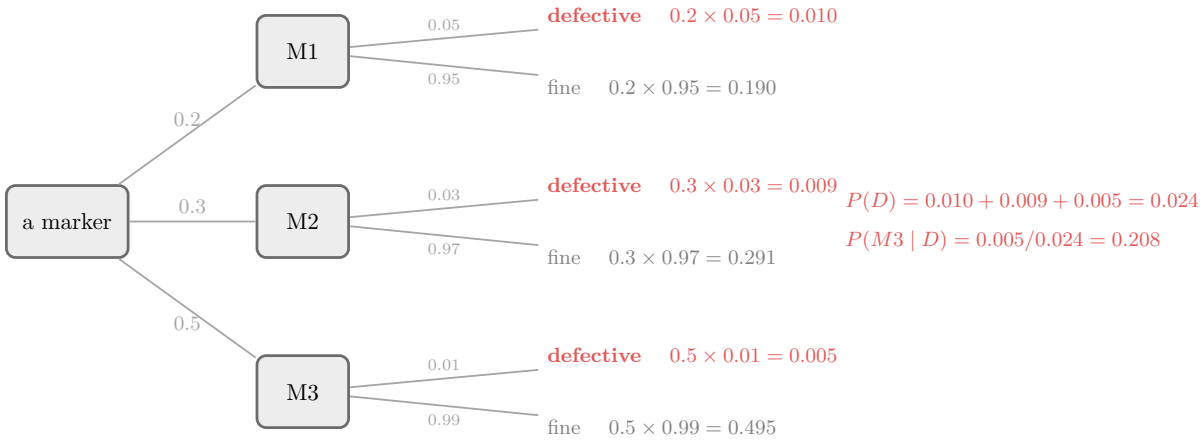


Figure 3: Probability tree: each path multiplies its branches

5. The answer

> **Key point:** Given a defect, the chance that M3 made the marker is only 0.208: M3 makes half the markers but only about a fifth of the defective ones.

$$P(M3 \mid D) = \frac{0.01 \times 0.5}{0.024}$$

$$= \frac{0.005}{0.024}$$

$$= 0.208$$

The same calculation for every machine:

Machine	Prior $P(M)$	**Joint probability** (G-986) $P(D \cap M)$	**Posterior** (G-1536) $P(M \mid D)$
M1	0.20	0.010	0.417
M2	0.30	0.009	0.375
M3	0.50	0.005	0.208

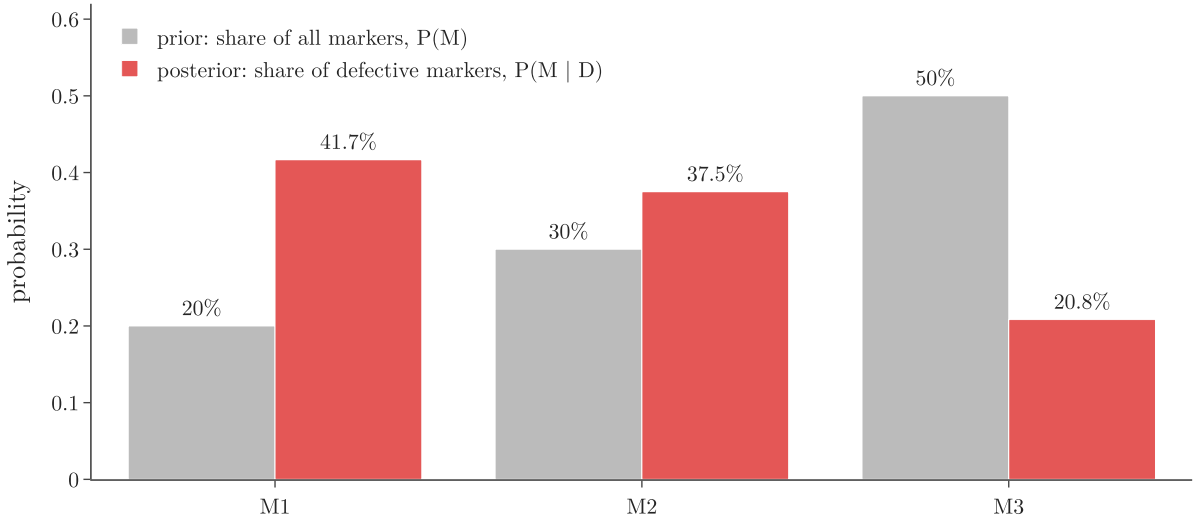


Figure 4: Share of all markers against share of defective markers

In Figure 4, each machine has two bars: its share of all markers and its share of the defective ones. Seeing that the marker is defective changes the picture completely. Before, M3 was the most likely source (50%). After, M3 is the least likely (21%), because M3 rarely makes defects.

An everyday picture: three cooks share a kitchen, and one dish comes out burnt. The cook who makes the most dishes is not the likely culprit if that cook almost never burns anything. M1 makes only 20% of markers but 42% of the defective ones.

The three posteriors add up to 1, as they must: the marker came from some machine. A simulation of a million markers gives $P(M3 \mid D) = 0.2095$, matching 0.208.

Python: The whole calculation.

```
prior = {"M1": 0.2, "M2": 0.3, "M3": 0.5}
p_defect = {"M1": 0.05, "M2": 0.03, "M3": 0.01}

joint = {m: prior[m] * p_defect[m] for m in prior}
p_D = sum(joint.values())                          # 0.024
{m: joint[m] / p_D for m in prior}                 # 0.417, 0.375, 0.208
```

6. Another way to see it: a medical test

Key point: A test does not give the final answer; it updates the prior. Posterior **odds** (G-1376) = prior odds × Bayes factor.

The marker problem has the same shape as a famous question about medical tests. The machines become "has cancer" and "does not have cancer"; "defective" becomes "the test is positive".

A screening test is given to 1000 women. What we know:

- **Prior:** 1 percent have breast cancer, so 10 women are sick and 990 are healthy.
- **Sensitivity** (G-1641; also called recall, the share of real positives found): the test is positive for 90 percent of the sick women: $P(+ \mid \text{cancer}) = 0.9$.
- **False positive rate** (G-749): the test is also positive for 9 percent of the healthy women: $P(+ \mid \text{healthy}) = 0.09$.

A woman tests positive. What is the chance that she has cancer? The test sounds "90 percent accurate", so many people answer 9 in 10. Count instead, exactly as with the markers:

1. **Sick and positive** (true positives):

$$10 \times 0.9 = 9 \text{ women}$$

2. **Healthy and positive** (false positives):

$$990 \times 0.09 = 89.1 \approx 89 \text{ women}$$

3. **All positives** (the evidence, by total probability):

$$9 + 89 = 98$$

4. **Posterior:** the share of positives who are sick.

$$P(\text{cancer} \mid +) = 9/98 \approx 0.092$$

That is about 1 in 11.

The healthy group is so large that its few false positives (89) swamp the true positives (9). This **posterior** (G-1536) is the **precision** (G-1547) of the test; in medicine it is called the positive predictive value.

The lesson: the test did not decide anything. The test moved the chance from 1 in 100 (the **prior** (G-1565)) to about 1 in 11 (the posterior). A test **updates** the prior.

6.1 The size of the update: the Bayes factor

> **Key point:** Bayes factor = P(evidence | yes) / P(evidence | no). Multiply the prior odds by it to get the **posterior** (G-1536) odds.

How strong is the update? A positive result is 10 times more likely for a sick woman than for a healthy one:

$$0.9/0.09 = 10$$

This ratio of the two **likelihoods** (G-1086) is the **Bayes factor** (G-2215).

The Bayes factor works on **odds** (G-1376): the number of "yes" cases to the number of "no" cases.

1. **Prior odds:** 10 sick to 990 healthy, that is $1 : 99$.
2. **Multiply by the Bayes factor.** These are the posterior odds.

$$1 \times 10 : 99 = 10 : 99$$

3. **Back to a probability:** 10 sick for every 99 healthy.

$$10/(10 + 99) = 0.092$$

The same answer as the count.

The rule is exact, and it follows from the count above: among the positives, the sick are scaled by 0.9 and the healthy by 0.09, so their ratio is multiplied by 0.9/0.09.

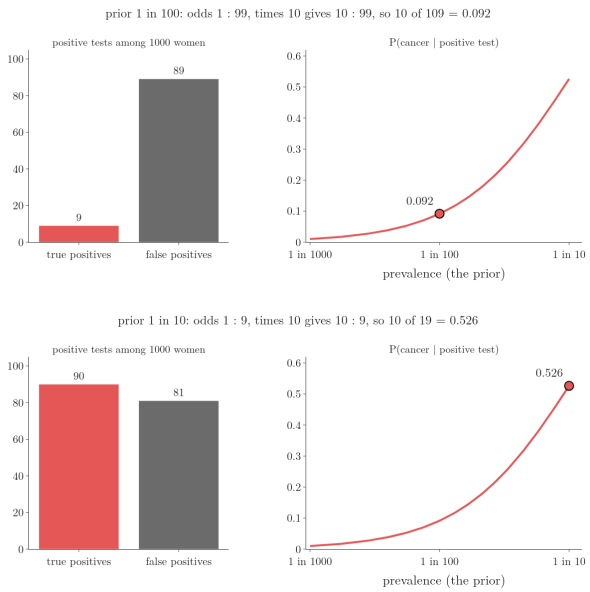


Figure 5: A test with sensitivity 0.9 and false positive rate 0.09 (Bayes factor 10), for priors from 1 in 1000 to 1 in 10. Left: expected true and false positives among 1000 women. Right: the posterior P(cancer | positive). The title gives the update in odds.

In Figure 5 the horizontal axis of the right panel is a log scale: 1 in 1000, 1 in 100 and 1 in 10 are equally spaced, each ten times the one before. Watch the same test give very different answers as the **prior** (G-1565) changes. At a prior of 1 in 1000 a positive result means about 1 percent; at 1 in 100, 9 percent; at 1 in 10, the odds 1 : 9 become 10 : 9, which is 53 percent. The test is the same each time; only the prior differs.

The Bayes factor also previews Naive Bayes: each feature multiplies the score of a class by its own likelihood, one factor per feature (the naive assumption).

7. The pattern for Naive Bayes

> **Key point:** For each class: prior × **likelihood** (G-1086). Divide by their total. The largest posterior wins.

A Naive Bayes classifier follows exactly this procedure. The machines become the classes, the values of the **target** (G-1949; the output we predict). "Defective" becomes the observed **features** (G-772; the input variables, one column each of the data table):

1. for every class, multiply its **prior** (G-1565) by the likelihood of the observed features;
2. divide each product by their sum (the evidence) to get **posteriors** (G-1536) that add to 1;
3. predict the class with the largest posterior.

Since the evidence is the same for every class, step 2 does not change which class wins; it only turns the scores into probabilities.

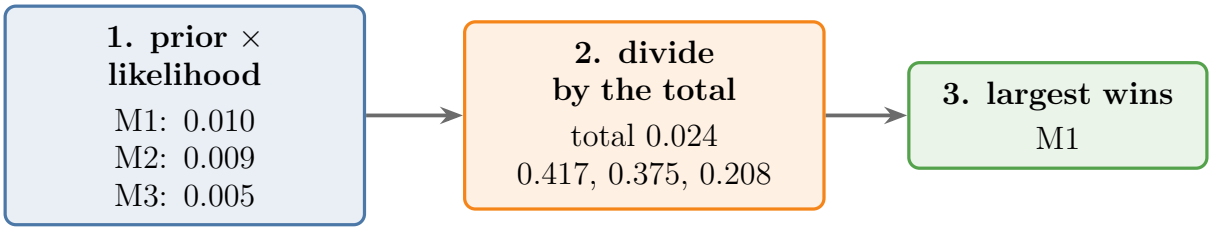


Figure 6: The three Naive Bayes steps on the marker problem

Figure 6 runs the three steps on the marker numbers: M1 has the largest score both before and after dividing by 0.024.

8. Summary

- Priors: production shares; **likelihoods** (G-1086): defect rates; posterior: which machine, given a defect. Bayes' theorem turns the chance of a defect given the machine into the chance of the machine given a defect.
- **Law of total probability** (G-1053): $P(D) = \sum_i P(D \mid M_i)P(M_i)$, because a defective marker came from exactly one machine, so the three cases add. The sign $\sum_i$ means "add the terms for $i = 1, 2, 3$", one per machine (0.010, 0.009 and 0.005, which add to 0.024).
- Answer: $P(M3 \mid D) = 0.208$; M1 is the most likely source (0.417), because M3 rarely makes defects: it makes half the markers but only about a fifth of the defective ones.
- Medical test: prior 1 in 100, sensitivity 0.9, false positive rate 0.09 give a posterior of only 0.092 (9 sick among 98 positives), because the healthy group is so large that its few false positives swamp the true positives. A test updates the prior; it does not decide. Posterior **odds** (G-1376) = prior odds × Bayes factor (section 6.1).
- Naive Bayes follows the same steps: prior × likelihood for each class, divide by the evidence, and predict the class with the largest **posterior** (G-1536).

9. Sources

Built from

- CampusX, "Naive Bayes Classifier | Part 5 | Problem based upon Bayes Theorem", YouTube, https://www.youtube.com/watch?v=aAEHjXDHtbE
- Sanderson, G. (3Blue1Brown), "The medical test paradox, and redesigning Bayes' rule", YouTube, https://www.youtube.com/watch?v=lG4VkPoG3ko (the 1000-women example, the test as an update, the Bayes factor and the odds form; Figure 5 is our own chart of those numbers)

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Law of total probability (G-1053)	The rule for the overall probability of an event: add its probability under each case, weighted by how likely that case is, $P(B) = \sum_i P(B \mid A_i)P(A_i)$, when the cases A_i are mutually exclusive and cover every possibility.

Term	Meaning
Probability tree (G-1573)	A diagram that draws each stage of a problem as branches labelled with probabilities; multiplying along a path gives that path's probability, and adding paths gives a total probability.
Bayes factor (G-2215)	A number that says how strongly the evidence favours yes over no: how many times more likely the evidence is if the answer is yes than if it is no, $P(\text{evidence} \mid \text{yes})/P(\text{evidence} \mid \text{no})$. Multiplying the prior odds by it gives the posterior odds.
Prior (G-1565)	The probability of an event before any evidence is seen.
Likelihood (G-1086)	How probable the observed data is under given parameter values; read as a function of the parameters with the data fixed.
Bayes' theorem (G-269)	The rule that reverses a conditional probability: from $P(B \mid A)$ it gives $P(A \mid B)$, so a belief about A can be updated after seeing B; $P(A \mid B) = P(B \mid A)P(A)/P(B)$.
Joint probability (G-986)	The probability that two events happen together, $P(A \cap B)$.
Posterior (G-1536)	The probability of an event after the evidence is taken into account.
Odds (G-1376)	How often an event happens divided by how often it does not, e.g. 5 placed to 3 not placed is 5/3; unlike a probability, odds have no upper limit, so evidence updates them by one multiplication, and their log can be predicted with a straight line.
Recall (sensitivity) (G-1641)	Of all items that really are positive, the fraction the model found.
False positive rate (FPR) (G-749)	The fraction of real negatives the model wrongly flags.
Precision (G-1547)	Of all items predicted positive, the fraction that really are positive.
Target (G-1949)	The output column we predict, such as the class; also called the label.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).